

## QAMISHLI, Syrian Arab Republic

WMO: 400010

Lat: 37.021N

Lon: 41.191E

Elev: 451

StdP: 96.02

Time Zone: 2.00 (SYR)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -1.4       | 0.1 | -11.8                         | 1.4 | 6.3  | -9.6 | 1.7 | 7.7  | 13.3                  | 8.6  | 11.7 | 8.1  | 3.9                   | 0    | 0.477 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.4                   | 43.1            | 19.7 | 42.0 | 19.4 | 40.8 | 19.3 | 22.2                | 37.2 | 21.2 | 36.8 | 20.5 | 36.6 | 4.5                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.2                            | 12.9 | 28.7 | 15.7 | 11.8 | 27.1 | 14.6 | 11.0 | 26.0 | 67.4          | 36.8 | 63.8 | 36.8 | 61.0 | 36.6 | 27.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.7              | 9.5  | 8.3 | -3.5                       | 45.0 | 1.9                | 1.7 | -4.8                                               | 46.2 | -5.9       | 47.1 | -7.0       | 48.1 | -8.4       | 49.3 |
|                   |      |     | -4.7                       | 23.8 | 1.8                | 1.8 | -6.0                                               | 25.0 | -7.1       | 26.1 | -8.1       | 27.0 | -9.4       | 28.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 19.8  | 6.9   | 8.4   | 12.2  | 16.5  | 23.2  | 29.8  | 32.9  | 32.5  | 27.8  | 22.5  | 14.4  | 9.2  | (6)  |
| (7)  |                                                                   | DBStd       | 9.76  | 2.75  | 3.18  | 3.68  | 3.76  | 3.65  | 2.94  | 2.53  | 2.46  | 2.90  | 3.73  | 3.58  | 3.15 | (7)  |
| (8)  |                                                                   | HDD10.0     | 238   | 99    | 61    | 18    | 2     | 0     | 0     | 0     | 0     | 0     | 7     | 52    | (8)  |      |
| (9)  |                                                                   | HDD18.3     | 1321  | 353   | 277   | 193   | 78    | 5     | 0     | 0     | 0     | 8     | 122   | 283   | (9)  |      |
| (10) |                                                                   | CDD10.0     | 3801  | 4     | 17    | 87    | 197   | 409   | 594   | 708   | 697   | 534   | 388   | 140   | 27   | (10) |
| (11) |                                                                   | CDD18.3     | 1843  | 0     | 0     | 3     | 24    | 156   | 344   | 450   | 438   | 284   | 138   | 6     | 0    | (11) |
| (12) |                                                                   | CDH23.3     | 26911 | 0     | 0     | 21    | 258   | 1900  | 5130  | 7407  | 7069  | 3721  | 1358  | 44    | 3    | (12) |
| (13) |                                                                   | CDH26.7     | 17021 | 0     | 0     | 4     | 70    | 906   | 3258  | 5155  | 4862  | 2199  | 564   | 2     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 3.7   | 3.9   | 3.8   | 3.8   | 3.5   | 3.7   | 4.1   | 3.9   | 3.4   | 3.3   | 3.3   | 3.1   | 4.0  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 419   | 92    | 66    | 76    | 64    | 27    | 3     | 0     | 1     | 22    | 41    | 69    | (15) |      |
| (16) |                                                                   | PrecMax     | 728   | 399   | 159   | 468   | 266   | 171   | 30    | 6     | 2     | 15    | 89    | 153   | 228  | (16) |
| (17) |                                                                   | PrecMin     | 219   | 4     | 12    | 9     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | (17) |      |
| (18) |                                                                   | PrecStd     | 134   | 73    | 41    | 75    | 50    | 39    | 6     | 1     | 0     | 3     | 24    | 31    | 51   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | DB          | 17.0  | 19.1  | 25.9  | 32.2  | 37.4  | 42.8  | 45.0  | 44.9  | 41.1  | 35.3  | 26.4  | 22.7  | (19) |      |
| (20) |                                                                   | MCWB        | 9.3   | 10.5  | 12.7  | 15.3  | 17.5  | 18.7  | 19.8  | 20.0  | 18.0  | 16.7  | 13.3  | 11.2  | (20) |      |
| (21) |                                                                   | DB          | 14.3  | 17.2  | 23.0  | 28.7  | 35.1  | 40.8  | 43.4  | 43.0  | 39.1  | 33.5  | 24.7  | 18.2  | (21) |      |
| (22) |                                                                   | MCWB        | 8.2   | 10.0  | 12.7  | 15.5  | 17.1  | 18.9  | 19.9  | 19.6  | 18.1  | 16.6  | 12.6  | 9.7   | (22) |      |
| (23) |                                                                   | DB          | 12.9  | 15.6  | 21.1  | 26.7  | 33.2  | 39.2  | 42.2  | 41.9  | 37.2  | 31.8  | 22.8  | 16.2  | (23) |      |
| (24) |                                                                   | MCWB        | 7.5   | 9.1   | 12.0  | 15.0  | 16.8  | 18.4  | 19.6  | 19.6  | 17.9  | 16.4  | 12.4  | 9.3   | (24) |      |
| (25) |                                                                   | DB          | 11.4  | 13.9  | 19.1  | 24.4  | 31.6  | 37.9  | 41.0  | 40.8  | 35.8  | 29.9  | 20.9  | 14.3  | (25) |      |
| (26) |                                                                   | MCWB        | 7.2   | 8.3   | 11.4  | 14.3  | 16.6  | 18.1  | 19.3  | 19.5  | 17.6  | 15.7  | 11.8  | 8.5   | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | WB          | 11.0  | 12.6  | 16.4  | 19.1  | 21.2  | 23.5  | 23.0  | 23.2  | 23.2  | 19.1  | 16.1  | 12.2  | (27) |      |
| (28) |                                                                   | MCDWB       | 14.6  | 16.8  | 20.7  | 27.6  | 31.2  | 34.0  | 39.5  | 38.7  | 33.8  | 30.9  | 20.5  | 17.4  | (28) |      |
| (29) |                                                                   | WB          | 9.8   | 10.9  | 14.3  | 17.0  | 19.1  | 20.8  | 21.9  | 22.1  | 20.2  | 18.0  | 14.8  | 11.4  | (29) |      |
| (30) |                                                                   | MCDWB       | 12.3  | 15.1  | 20.1  | 23.6  | 29.2  | 35.0  | 39.0  | 36.9  | 33.3  | 29.2  | 20.0  | 15.6  | (30) |      |
| (31) |                                                                   | WB          | 8.9   | 10.0  | 13.1  | 16.0  | 18.2  | 19.6  | 21.1  | 21.2  | 19.1  | 17.2  | 13.9  | 10.7  | (31) |      |
| (32) |                                                                   | MCDWB       | 11.3  | 13.8  | 19.0  | 23.5  | 29.1  | 35.2  | 38.2  | 36.9  | 33.6  | 28.3  | 19.4  | 14.2  | (32) |      |
| (33) |                                                                   | WB          | 8.1   | 9.2   | 12.1  | 15.1  | 17.4  | 18.8  | 20.3  | 20.5  | 18.3  | 16.5  | 12.9  | 9.9   | (33) |      |
| (34) |                                                                   | MCDWB       | 10.6  | 12.9  | 18.1  | 22.7  | 28.6  | 35.8  | 38.0  | 36.8  | 33.5  | 27.1  | 19.1  | 13.1  | (34) |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 8.1   | 9.0   | 10.6  | 12.1  | 14.0  | 14.8  | 15.4  | 15.8  | 14.9  | 12.3  | 10.9  | 8.8   | (35) |      |
| (36) |                                                                   | 5% DB       | 10.6  | 11.2  | 12.9  | 14.5  | 15.6  | 15.5  | 16.0  | 16.3  | 16.2  | 13.9  | 13.0  | 11.8  | (36) |      |
| (37) |                                                                   | MCWBR       | 6.1   | 6.1   | 6.5   | 6.4   | 5.5   | 5.6   | 6.1   | 6.5   | 6.9   | 6.1   | 6.1   | 6.4   | (37) |      |
| (38) |                                                                   | 5% WB       | 7.2   | 9.2   | 11.2  | 12.1  | 13.7  | 14.4  | 14.8  | 15.4  | 14.6  | 12.0  | 10.1  | 8.3   | (38) |      |
| (39) |                                                                   | MCWBR       | 4.8   | 5.6   | 6.4   | 6.5   | 5.8   | 6.1   | 6.2   | 6.6   | 6.9   | 5.7   | 5.1   | 5.2   | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.373 | 0.403 | 0.433 | 0.497 | 0.477 | 0.422 | 0.419 | 0.415 | 0.391 | 0.439 | 0.376 | 0.370 | (40) |      |
| (41) |                                                                   | taud        | 2.185 | 2.104 | 2.042 | 1.895 | 1.982 | 2.134 | 2.158 | 2.166 | 2.233 | 2.062 | 2.215 | 2.198 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 786   | 810   | 824   | 791   | 815   | 860   | 860   | 855   | 857   | 760   | 777   | 759   | (42) |      |
| (43) |                                                                   | Edh at Noon | 110   | 135   | 157   | 193   | 180   | 154   | 150   | 145   | 128   | 139   | 106   | 102   | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 2.28  | 3.12  | 4.45  | 5.52  | 6.83  | 8.11  | 7.89  | 7.07  | 5.84  | 4.04  | 2.90  | 2.09  | (44) |      |
| (45) |                                                                   | RadStd      | 0.25  | 0.37  | 0.31  | 0.43  | 0.45  | 0.33  | 0.18  | 0.21  | 0.22  | 0.28  | 0.27  | 0.28  | (45) |      |

## Historical Trends

|      |                        | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |      |
|------|------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |      |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) |      |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A | (46) |
| (47) | Regional (3 neighbors) | N/A     | N/A    | N/A     | N/A   | -1.67 | N/A         | N/A     | N/A     | N/A     | +51 | (47) |

Nomenclature: See separate page